Natural Language Processing

Term Project: Arabic Bank Check Amount Extraction and Processing

Important Note: The check dataset is not public and it has a license. So, make sure you do not share with anyone or upload it on a public website.

Part-A [20 points]: Legal and Courtesy Amount Extraction

A bank check has legal amount and courtesy amount regions as shown in the figure below with green boxes:

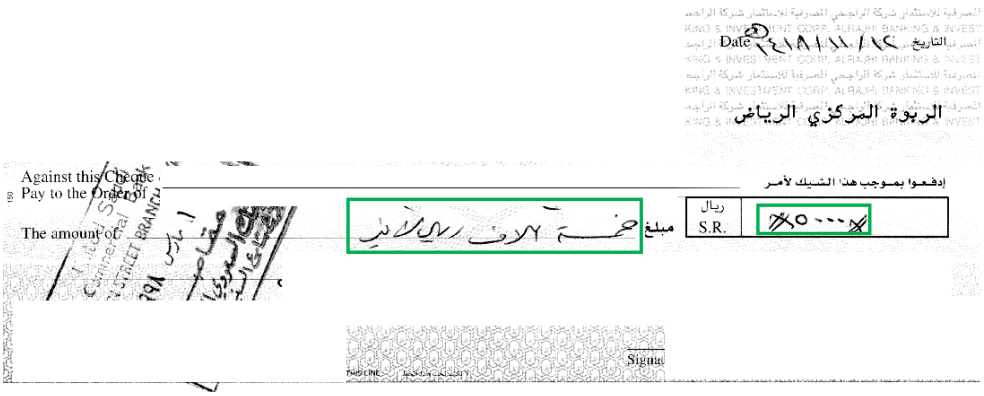

Figure 1: An example Arabic bank check image.

In this phase, your task is to train an object detection model to automatically extract the legal and courtesy amount regions from the bank check images. You need to train/validate your model using the dataset provided. Later your model will be evaluated on a test set to judge the final performance of the extraction system. Accuracy with IoU at different threshold (at 50%, 75% and 90%) and mean IoU will be used as evaluation metrics:

$$\text{Accuracy}_t = \frac{\text{Samples with IoU} \geq t}{\text{Total number of samples}}$$

$$\text{Mean IoU} = \frac{1}{N} \sum_{i=1}^{N} \text{IoU}_i$$

Input: Check Image

Output: {Check File Name} {Bounding box courtesy amount} {Bounding box legal amount}

<u>Part-B [35 points]: Courtesy Amount Recognition</u>

For this part of the project your task is to train a sequence model to automatically recognize the digit sequences in a courtesy amount. You will feed in the automatically extracted courtesy amount image patch from Part-A to a sequence model to decode the amount written on the image. For example, inputting the below courtesy amount region to the model should output **80000**

Figure 2: Courtesy amount (80000) image patch from a check.

You need to report the results using the following metrics:

1. **Accuracy at the digit level**. Accuracy at the digit level is computed using the following formula:

$$Accuracy\ (\%)\ =\ \left(1 - \frac{I\ + D + S}{N}\right) \times\ 100$$

   where I = number of insertions, D= number of deletions, S= number of substitutions, and N= total number of digits in an amount.

2. Percentage of amounts with no errors
3. Percentage of amounts with only one error
4. Percentage of amounts with two or more errors

Input: Courtesy Amount Image Patch

Output: {Check File Name} {Amount as a sequence of digits without any special characters}

Part-C [35 points]: Legal Amount Recognition

For this part of the project your task is to train a deep-learning model to recognize the legal amount. You will feed in the automatically extracted legal amount image patch from Part-A to a model to decode the amount written on the image. For example, inputting the below legal amount region to the model should output:

عشرة الاف ريال فقط لاغير

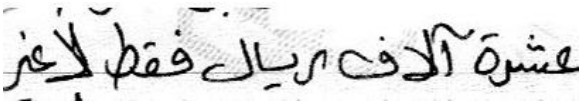

Figure 3: Legal amount image patch from a check.

You need to report the results using the following metrics:

1. **Error rate at the character (CER) and word level (WER)**. Accuracy at a token level is computed using the following formula:

$$CER|WER\ (\%)\ =\ \left(\frac{I\ +D+S}{N}\right)\ \times\ 100$$

   where I = number of insertions, D= number of deletions, S= number of substitutions, and N= total number of tokens in an amount.

Input: Legal Amount Image Patch

Output: {Check File Name} {Amount as a sequence of characters}

Part-D [10 points]: Final Verification

In this stage you convert the legal amount to digits using a rule-based script and then match with the courtesy amount. If they match then the courtesy amount is verified. If they don't match then the verification fails.

Extra (Bonus): You can mutually improve the recognized courtesy amount and legal amount and improve the results, further.

Submission: Code + Project Report**:**

You need to submit a project report (along with the code). The report should include the following sections:

- Problem definition
- Dataset
- Methodology
- Experiments and results
- Discussion including error analysis
- Conclusions including the limitations

The exact sections and the organization of the report is your choice but it should include the necessary information as listed above.
Consult the instructor for further clarifications, if needed.